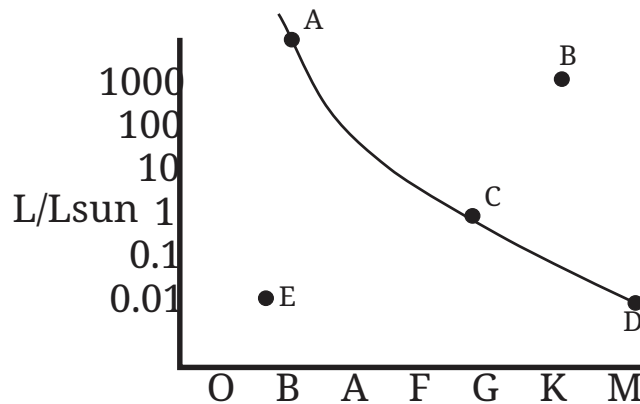




Star Ranking

Name: Connor Petri

- 1) The HR Diagram below shows the position of 5 stars.



- a) Rank the stars on the basis of temperature, hottest first.

EACBD

- b) Rank the stars on the basis of Luminosity, brightest first.

ABCED

- c) Rank the stars according to the radius, smallest first.

EDCAB

- d) Rank the Main Sequence stars according to mass, smallest first.

DCA

- e) Rank the Main Sequence stars according to lifetime of the stars, shortest first.

ACD

- 2) The table below shows the apparent magnitude (V), Absolute Magnitude (M), and (B-V) for 5 stars.

Star	V	M	(B-V)
A	5.5	-6.3	-0.2
B	2.5	1.4	0.1
C	-1.0	3.1	0.23
D	10.5	1.4	0.15
E	1.3	3.1	0.25

- a) Rank the stars according to the brightness in the sky, brightest first.

C E B A D

- b) Rank the stars according to the Luminosity, most luminous first.

A B=D C = D

- c) Rank the stars according to temperature, hottest first.

A B D C E

- d) Rank the stars according to distance, closest first (Remember, V-M is the distance modulus, M is the magnitude at 33 light years away)!!

C E B D A

3) Below is a table of different phases of the evolution of a star.

A	Planetary Nebula	G	Black Hole
B	G Type Main Sequence	H	Cepheid Variable
C	Red Giant	I	Supernova Type II
D	Molecular Cloud	J	O Type Main Sequence
E	Neutron Star	K	White Dwarf
F	Supernova Type I	L	B Type Main Sequence

a) Rank the table in order for the sun. Use only the phases that apply.

DBCAK

b) Rank the table in order for the highest mass (O-type) star. Use only the phases that apply.

DOIG

c) Rank the table in order for a high mass (B-type) star. Use only the phases that apply.

DLIE

d) Rank the table in order of a G- type star with a K-type star orbiting very close to it.

DBCAKFE

4) The table below gives the apparent magnitude, m, and period for 3 Cepheid variables.

Star	m	Period (days)
A	4.2	60
B	1.5	15
C	4.2	27

a) Rank the stars according to the Luminosity of the star, largest first

A C B

b) Rank the stars according to distance, closest first.

B C A

The table below list different objects.

A	California	E	San Jose
B	Neutron Star	F	You
C	Red Giant	G	Jupiter
D	White Dwarf	H	Sun

Rank these objects according to the radius, smallest first.

FBEADGHC